

Ayon Sarkar

+91-81002-11809 | ayon380@icloud.com | [GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Medium](#)

EXPERIENCE

- **Nucleus AI**
Software Development Engineer 1
 - Scaled asynchronous cloud microservices across **300+ concurrent AWS Lambdas** in **Python and Node.js**, handling high-volume event payloads with retries and dead-letter recovery.
 - Implemented **Apache Kafka** distributed event streaming to decouple and stream events across aesthetic scoring pipelines and downstream data transformation stages.
 - Engineered **Spectra**, an internal Next.js S3 explorer and Parquet visualizer leveraging **DuckDB (Wasm)** for client-side SQL queries, telemetry monitoring, and secure AWS Pre-signed URLs.
 - Built responsive web platforms using **Next.js Server-Side Rendering (SSR)** with dynamic MDX rendering, boosting organic discovery and cutting FCP by 45%.
 - **Nucleus AI**
Frontend Intern
 - Engineered responsive product landing pages and mobile interfaces with dynamic CSS animations, achieving sub-second load times and 100/100 Core Web Vitals.
 - Built modular, reusable UI component systems using **React, TypeScript, and Tailwind CSS**, conducting automated testing with **Jest** in an Agile TDD workflow.
 - Integrated modern AI-assisted workflows (**Claude Code, GitHub Copilot**) for rapid prototyping, automated code generation, and prompt evaluation.
 - **Agumentik Group**
Software Developer Intern
 - Delivered **10+ production REST endpoints** maintaining **sub-50ms p95 latencies** and secure JWT authentication using **Node.js, Express, Sequelize, and PostgreSQL/MySQL**.
 - Designed relational database schemas, connection pooling, and multi-tier caching with **Redis**, reducing database query latency by 35%.

Remote (India/US Team)
August 2025 – April 2026

Remote (India/US Team)
August 2024 – July 2025

Remote (India)
June 2023 – August 2023

SKILLS

Languages:	TypeScript, JavaScript (ES6+), Python, Node.js, SQL, HTML5/CSS3, Go, Bash/Shell
Frontend & Mobile:	React, Next.js, React Native, Progressive Web Apps (PWA), Zustand, Tailwind CSS, WebSockets
Backend & Cloud:	Node.js, Express, RESTful APIs, PostgreSQL, Supabase, Redis, AWS (Lambda, S3, SQS), Docker, Sequelize ORM
AI & Engineering:	LLM APIs (Gemini/OpenAI), AI Tracing / Observability, Jest (TDD), Git, CI/CD Pipelines, Agile SDLC

PROJECTS

- **Aptinova — AI-Powered Hiring & Assessment Platform**
Tools: Next.js 15, React 19, TypeScript, Tailwind CSS, Zustand, Express, PostgreSQL, Gemini AI, WebSocket, WebAuthn
 - Built a full-stack AI hiring platform with 3 distinct user roles (candidate, HR, HR Manager), dashboards, applicant tracking, analytics, and interview scheduling (100/100 Lighthouse score).
 - Engineered a **Google Gemini AI** structured extraction pipeline parsing and ranking unstructured candidate documents into schema-validated profiles with source attribution; built an assessment module with 8+ real-time signals.
 - Implemented **Passkey / WebAuthn**, OAuth, JWT authorization between services, Razorpay subscription payments, and Monaco code editor.
 - **Muse — Full-Stack Social Media Platform**
Tools: Next.js 14, React 18, TypeScript, Tailwind CSS, Zustand, Firebase, WebRTC, Next PWA
 - Built a full-stack social media platform with **Next.js 14, React 18, and Zustand**, featuring infinite-scroll feeds, Reels with in-app sharing, multimedia posts, and nested comment replies.
 - Implemented real-time direct messaging with active user presence, custom chat theming, in-chat media/file sharing, and local P2P audio/video calling via **WebRTC**.
 - Engineered social graph architecture with public/private accounts, follow/follower graphs, and responsive Progressive Web App (Next PWA) with offline caching.

October 2024 – Present
[\[Live\]](#) [\[Code\]](#)

January 2024 – Present
[\[Live\]](#)

ACHIEVEMENTS & CERTIFICATIONS

- **Juspay Hiring Challenge 2024** — Ranked **top 500 out of 266,000+ contestants** (top 0.2%) in national systems contest. 2024
- **IEEE Publication** — [Enhanced Environmental Perception for Visually Impaired IEEE Xplore](#). 2024
- **IEEE Technical Reviewer** — Appointed reviewer for the **4th IEEE IATMSI-2026** conference. [\[Certificate\]](#) 2026

EDUCATION

- **SRM University AP**
B.Tech in Computer Science — Cloud Computing Specialisation | **CGPA: 8.2/10**

Amaravati, India
August 2021 – May 2025